Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner specializing in fullstack architectures, cloud microservices, and AI powered data platforms. Demonstrated track record at Parspec and Terminal Trends building responsive web applications, optimizing database queries, and engineering backend APIs across Python, Java, Go, and TypeScript.

PROJECTS

OmniTrack AI | Full Stack Financial Intelligence & Analytics Platform GitHub

- Built a responsive fullstack web dashboard using React, Node.js, and PostgreSQL to visualize financial transaction metrics and structured customer insights.
- Integrated Generative AI pipelines and Qdrant vector search to enable automated document summarization and sub 100ms context retrieval.
- Improved application responsiveness and throughput by 35 percent by implementing asynchronous API endpoints and optimized state management.

AgenticFlow | Real-Time Workflow Orchestration Engine GitHub

- Developed a scalable microservice orchestration platform in Go and Python to automate multi step background data processing tasks.
- Increased system throughput by 40 percent by designing asynchronous RESTful APIs, WebSocket event streams, and transactional database models in PostgreSQL.
- Applied Agile development methods, automated unit testing, and continuous integration pipelines to guarantee code quality and zero downtime deployments.

Low Latency Search Core | High-Throughput Data Processing Engine GitHub

- Constructed a high performance analytical indexing and search core using C++ and Java to query millions of structured records in real time.
- Reduced query execution latency by 35 percent using memory mapping, custom inverted indexes, and low level cache optimization techniques.
- Containerized deployment workflows using Docker and integrated comprehensive automated test suites to ensure robust backend maintainability.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

Frontend & Mobile: React, JavaScript, TypeScript, HTML5, CSS3, Responsive Web Design, REST API Integration, Redux
Backend & Frameworks: Python, Java, Go, C++, Node.js, Express, FastAPI, Spring Boot, RESTful APIs, Microservices, GraphQL
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Vector Databases (Qdrant), Redis, Docker, Kubernetes, AWS, Git, Linux
Engineering & AI Concepts: Object Oriented Design, Agile Methodologies, Generative AI Tools, System Design, Unit Testing, CI CD, Design for Delight

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.